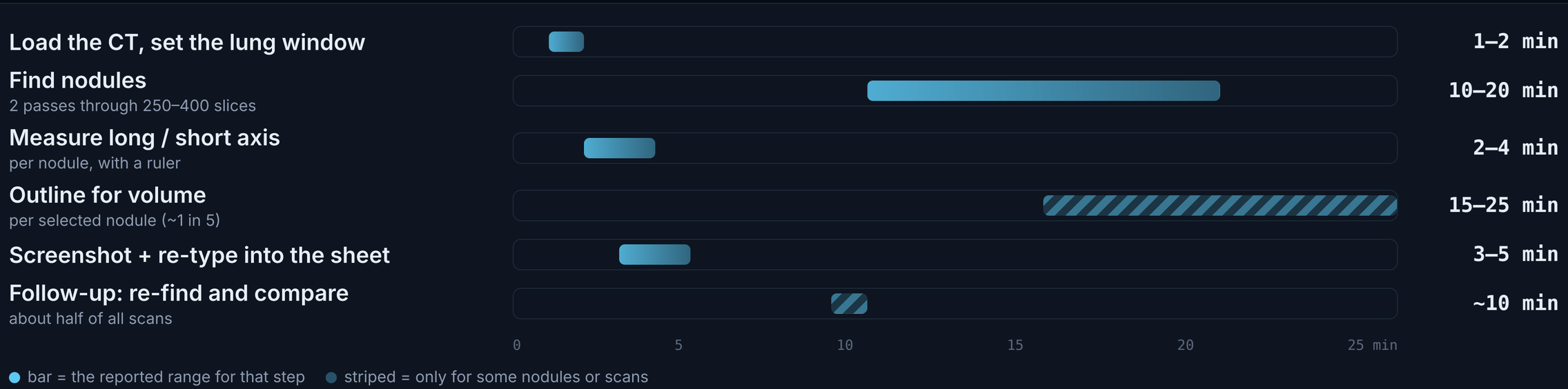
From manual CT review to AI-assisted research

Daniel's current baseline scan, step by step: his own time estimates, not measured by us.



TYPICAL BASELINE SCAN

25–40 min

Customer-reported. Naomi's target is about **50 % less**.

Can we cut this workload and still keep the researcher in control of every number?

Daniel's own constraint: the AI proposes, he reviews, fixes the edges, adds what was missed, and approves.

Let AI propose the findings



1 The model runs on the whole CT

TotalSegmentator 2.18.0, `lung_nodules`, offline on a MacBook. About 100 s per scan (measured), before the researcher sits down. Nothing tells it where to look.

2 Each 3-D piece becomes a candidate

Pieces under 5 mm³ are hidden. Each candidate is outlined in 3-D on every slice.

3 Measured in the browser

Long and short axis on the largest slice, volume from the voxels. A lobe hint comes from the AI's lung-lobe map.

THIS SCAN'S WORKLIST

AI N1	65.5 × 49.3 mm · 79,677 mm³ · RUL
AI N2	4.4 × 2.6 mm · LLL
AI N3	2.7 × 1.6 mm · RUL

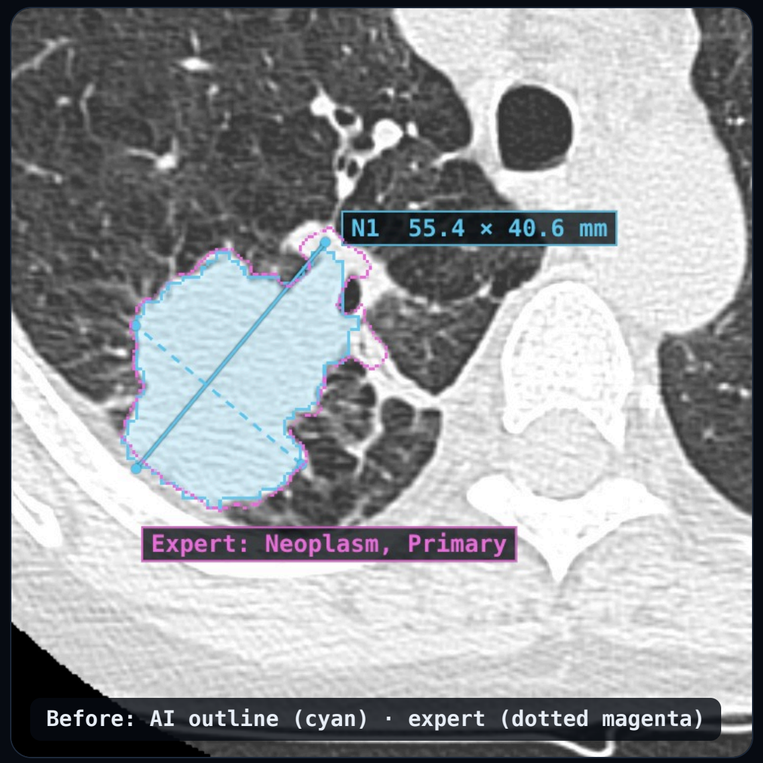
● every candidate starts "To review"

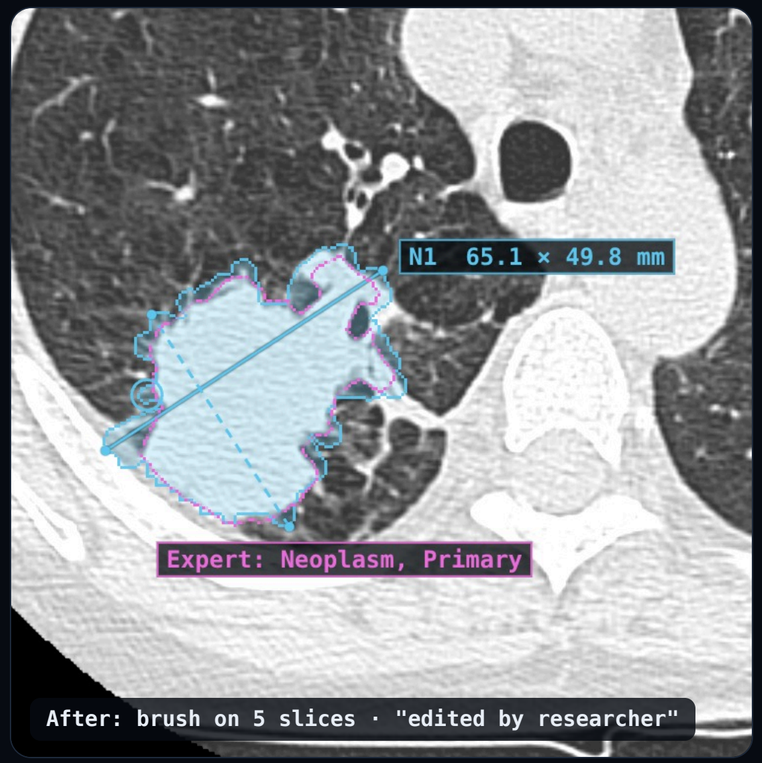
nothing is exported until it is accepted

Data: RIDER Lung CT (TCIA, CC BY) · AI file: TotalSegmentator 2.18.0 SEG · values read from the app on 20 Oct 2026

Evidence register E6, E20

Review. Correct. Approve.





FOLLOW-UP SCAN, N1	LONG AXIS	SHORT AXIS	VOLUME	DICE VS EXPERT
AI, as proposed	55.4 mm	40.6 mm	43.76 mL	0.87
After correction	65.1 mm	49.8 mm	46.79 mL	0.87
Expert (RIDER-LungCT-Seg)	59.8 mm	44.3 mm	54.99 mL	—

A **Accept or reject**
One key each. `J` / `K` moves through the list. A rejection can be undone.

B **Fix the edge**
`B` brush, `E` erase, `⌘Z` undo. Axes and volume are re-measured after every stroke.

S **Add what the AI missed**
`S` + click to outline and measure it. It is logged as added by the researcher.

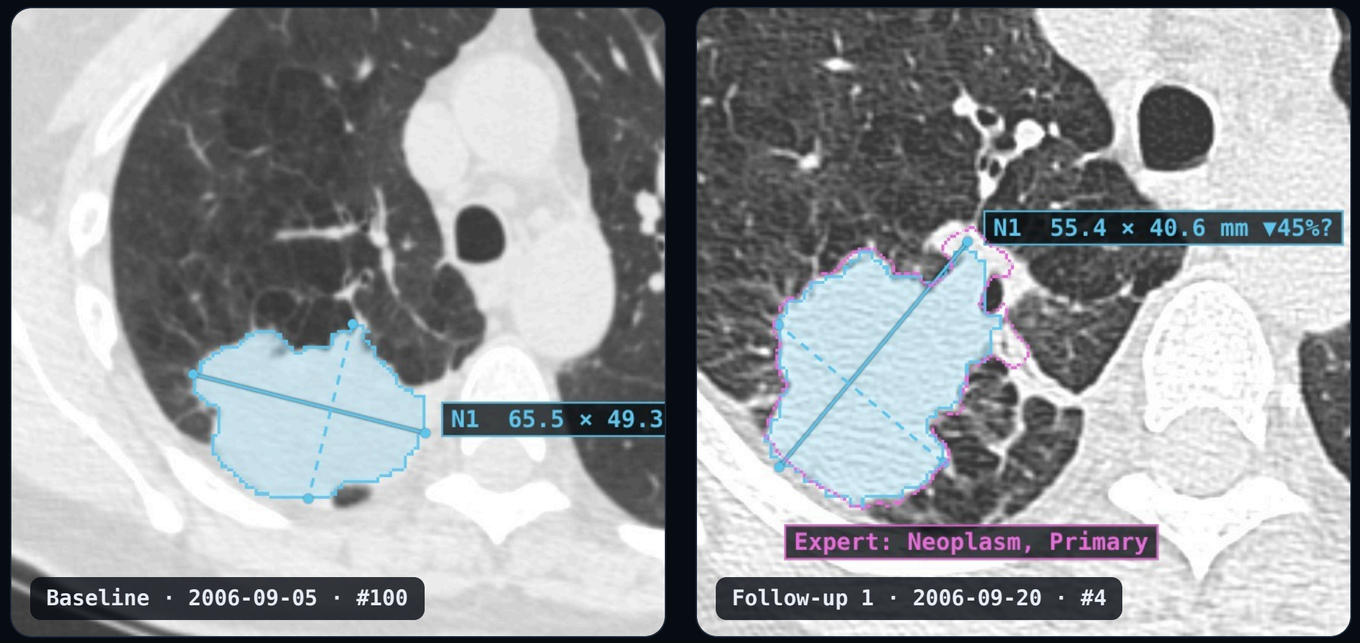
WHAT THIS EXAMPLE SHOWS

The volume moved toward the expert's (43.8 → 46.8 of 55.0 mL), and the change is tracked. One stroke at the left edge **overshot the long axis** (65.1 vs 59.8 mm). Undo is one key away, and correcting 5 of about 40 slices closed only part of the gap.

The strokes were made by our test script, not by a researcher, so this example tells us nothing about correction time.

RIDER-1129164940, 2006-09-20 scan · expert = RIDER-LungCT-Seg MANUAL segment · values read from the app after the strokes

Understand what changed



N1, AI OUTLINE ON BOTH SCANS	BASELINE	FOLLOW-UP	CHANGE
Long axis	65.5 mm	55.4 mm	−10.1 mm (−15.5 %)
Short axis	49.3 mm	40.6 mm	−8.7 mm (−17.6 %)
Volume	79,677 mm³	43,760 mm³	−35,917 mm³ (−45.1 %)

MATCHING STATUS

↔ ▼45% vs Baseline? suggested · confirmed by researcher (A / M)

Every follow-up finding is paired automatically with the nearest earlier finding (one-to-one, earlier scans only, ≤ 50 mm). Here the distance was 38 mm. The "?" stays until the researcher confirms the pair.

READ THIS CHANGE WITH CARE

- The two scans were reconstructed differently (**STANDARD vs LUNG kernel**), which by itself changes measured volume.
- The baseline has no expert outline. On the follow-up, the AI volume is 20 % below the expert's.
- Two scans of the same day differ by a median **9.5 % (our click) to 17 % (expert)** in volume, over 28 patients.

→ The app computes the change. Whether the tumour really shrank is the researcher's call.

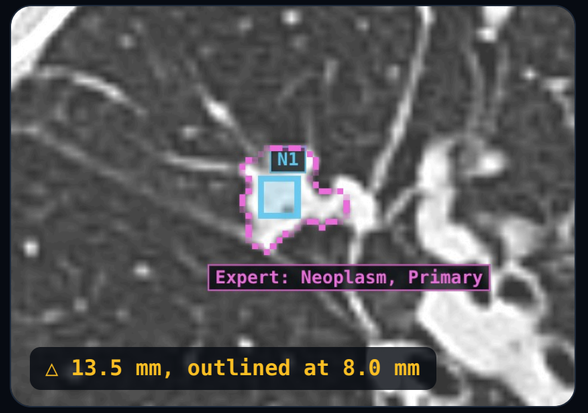
It saves the re-find and the side-by-side arithmetic, about **10 min per follow-up** by Daniel's estimate.

Dates, series and values read from the DICOM headers and the app · change = (follow-up − baseline) ÷ baseline

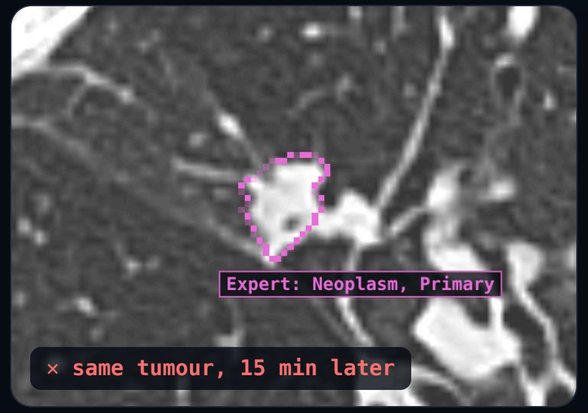
Testing where researchers lose the most time



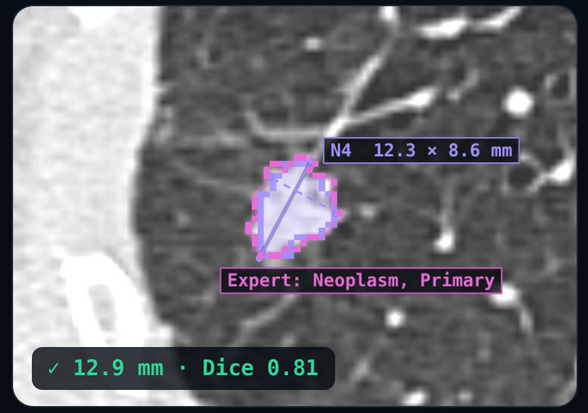
LIDC-IDRI-0075 · 4 radiologists outlined it (magenta). The AI put nothing there.



RIDER-1801720707 · found, but volume −83 %. Must be redrawn before use.



Repeat scan of the same patient: missed. The researcher adds it with **S** + click.



RIDER-2799584460 · AI (violet) on the expert outline. Cost: 15 other candidates to reject.

LIDC-IDRI · 15 SCANS · 35 NODULES OUTLINED BY ≥ 3 OF 4 RADIOLOGISTS

READER-RATED TYPE	NODULES	FOUND	DICE (FOUND)	VOLUME
Ground-glass	5	0	–	–
Part-solid	3	1	0.21	−88 %
Solid (6.0–42.6 mm)	27	26	0.60	−55 %

Radiologists agree with their own consensus at Dice 0.87. Volume is the median AI error vs the consensus. Extra candidates: median 1 per scan.

WHAT THIS MEANS FOR DANIEL

- **Ground-glass: the AI does not help today.** Keep searching for them by hand and add each one with **S** or the brush.
- **Small solid: found, but outlined too small.** Solid ≤ 10 mm: 15 of 16 found. All 27 nodules the AI found came out at least 26 % too small. Redraw the edge before using a volume.
- RIDER agrees: tumours ≤ 30 mm found 15/18; all 11 of the smallest that were found came out 26–97 % too small.

Real app screenshots · LIDC-IDRI (TCIA, CC BY 3.0) + RIDER-LungCT-Seg · AI = TotalSegmentator 2.18.0, run on an M3 Pro, median 97 s/scan

Does it actually save time?

What we measured, and what we have not measured yet. The 50 % target is shown as a goal, not as a result.

MEASURED · RIDER, 59 SCANS

53/59

known tumour found (≥ 1 voxel overlap) · 90 % · 29/31 patients · stricter Dice ≥ 0.5: 43/59

MEASURED · 53 HITS

0.76

median Dice vs expert outline [IQR 0.59–0.86] · long axis error 5.5 mm · volume error 26 %

MEASURED · NOT SCORABLE

3/scan

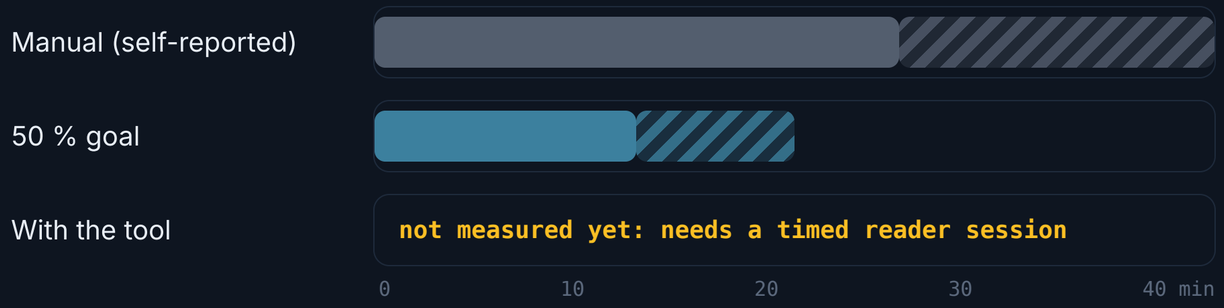
other candidates (median; up to 30). Only one tumour is labelled per scan, so these may be false positives or real nodules.

MEASURED · LIDC, 15 SCANS

97 s

median AI time per scan on a MacBook (M3 Pro), unattended, 69–129 s. RIDER file timestamps agree (~100 s).

RESEARCHER TIME PER BASELINE SCAN



LIMITS OF THIS EVENCE

- RIDER (59 scans) is mostly large solid tumours; LIDC adds 35 nodules on 15 scans, only 8 of them sub-solid
- False positives: only an upper bound (LIDC: median 1 per scan); not scorable on RIDER
- No researcher has been timed yet, and "manual" is Daniel's own estimate
- AI timing comes from one laptop. App timing comes from a slow test machine

MACHINE TIME (MEASURED, 3 RUNS)

Open sample → AI worklist ready	11.5–12.3 s
Accept (A) → saved, Type/Lobe filled	0.2–0.8 s
Switch to follow-up → pairs shown	8.6–9.1 s
Export CSV → file written	0.1–0.3 s

Headless browser with software graphics, slower than a laptop. Researcher actions, corrections and pauses are logged by the app for the timed session.

SECOND DATASET (LIDC-IDRI, MEASURED)

27 of 35 radiologist-confirmed nodules found (77 %): solid 26/27, part-solid 1/3, **ground-glass 0/5**. Median Dice 0.59 (radiologists vs consensus: 0.87). Every outline the AI produced was too small.

Ground-glass: 0 of 5 found on LIDC (slide 5) · no researcher has been timed yet

What would make this trustworthy enough to use?

WHAT YOU TOLD US

> "The workflow of AI-assisted finding followed by my manual review and correction is exactly what I need … Before I can call this a time-saver, we'll need to time the full process … I'd also like to see how it handles those smaller, hazier nodules."

Daniel Reyes, Slack. This is a written statement from a simulated stakeholder, not an observation of him using the tool.

WHERE WE STAND ON YOUR THREE POINTS

Review + correction	built; correction re-measures and is tracked (slide 3)
Full-process timing	instrumented; **no timed session yet** (slide 6)
Small / hazy nodules	tested on LIDC + RIDER: solid found but outlined too small; **ground-glass 0 of 5** (slide 5)

1. Would this review-and-correct flow fit how you work through a scan?
2. Which failure would make you stop trusting it: a missed nodule, a wrong edge, or a wrong pair?
3. Do the small and ground-glass results change whether you would use it?
4. What is the one remaining blocker to using it as a research tool?

Proposed next step: a 15–20 minute timed session on 10 public scans, manual vs AI-assisted.

Reply in the thread or comment on any slide. Every number here links to the evidence register.